Girls Build Bots — Brownie Cybersecurity Safeguards

Badge-aligned workshop guide for robotics-team-led outreach

Level: Brownies (Grades 2–3) · **Badge:** Cybersecurity — Cybersecurity Safeguards **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser games and simple props the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: being safe and smart online

The Cybersecurity series builds digital street-smarts: understand how information travels (Basics), **protect it** (Safeguards), and think critically to solve problems (Investigator). Keep it concrete and friendly — never scary.

Vocabulary: password, protect, safeguard, secret, strong vs. weak, permission, ask a grown-up.

Cybersecurity Safeguards (75–90 minutes)

Badge theme: Practicing the safeguards that keep us safe online — strong passwords, permission, and what (not) to share.

Materials

- A “treasure box” with a simple lock/latch (a prop for the password idea)
- **Make-believe** password cards to sort strong vs. weak (never real passwords)
- Devices with **Interland** (beinternetawesome.withgoogle.com) — the “Tower of Treasure” realm is about strong passwords
- “Ask a grown-up?” scenario cards (a pop-up appears, a stranger messages you, a game asks for your name)

Activity 1 — The Secret Latch (Passwords) (15 min)

A password is like the latch on a treasure box — it keeps your things safe. Show the box: a weak latch (anyone can open it) vs. a strong one. Sort **make-believe** password cards into strong (long, mixed, silly-but-memorable) vs. weak (“1234,” “password”). **Never use anyone’s real passwords** — invented examples only.

Activity 2 — Play Interland: Tower of Treasure (30 min)

Scouts play the password-themed realm, building strong “keys” through the game. Mentors connect it back to the treasure-box idea: long and unexpected is strong.

Activity 3 — Ask a Grown-Up (15 min)

Scenario cards: a pop-up says “you won a prize!”, a game asks for your full name, someone you don’t know sends a message. For each, Scouts vote: safe, or **stop and ask a trusted grown-up?** The safeguard here is knowing when to pause and ask.

Activity 4 — My Safeguards (5 min) + Wrap (5 min)

Each Scout names one way she’ll keep her information safe (a strong password, asking a grown-up). Thumbs up if you know what makes a password strong.

> **Robot moment (optional):** Share for 60 seconds how the team uses strong passwords to protect its robot code and its accounts — the same safeguards, for a robot.

Completion Checklist — Cybersecurity Safeguards

By the end of the session, to earn the badge, each Scout can: - [ ] Explain what a password does and what makes one strong vs. weak - [ ] Build or identify a strong (make-believe) password - [ ] Name a situation where she should stop and ask a trusted grown-up - [ ] Name one safeguard she’ll use to protect her information

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Mentor Guide (for the robotics team)

- **You don’t need a robot for this.** A prop box and free browser games are enough. The “robot moment” box is optional.
- **Never handle real passwords.** All password examples are invented. Say this out loud so Scouts learn not to share real ones — even with you.
- **“Ask a grown-up” is a superpower, not a failure.** Frame pausing-to-ask as the smart, strong choice.
- **Keep it friendly, never scary.** Confidence and habits, not fear.
- **Watch the clock.** Brownies fade after ~25 minutes; keep segments moving.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit “Girls Build Bots” anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.

3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.